Review

Consider a differential equation $\frac{dy}{dt} = g(t, y)$.

Question What does it mean for a function $y = y(t)$ to be a *solution* of the equation?

Question Check if $y = e^{x^2/2}$ is a solution of $\frac{d^2y}{dx^2} = x\frac{dy}{dx} + y$.

Question Check if $y = 2$ is a solution of $\frac{dy}{dx} = y^2 - 4$.

Equilibrium Solutions

Definition An *equilibrium solution* is a solution that is constant for all values of t. That is, an equilibrium solution is a *value of* y that is a solution.

Example In the second question above, $y = 2$ is an equilibrium solution of $\frac{dy}{dx} = y^2 - 4$. What is another one?

Question Since equilibria are constant functions, for any equilibrium solution $y(t)$,

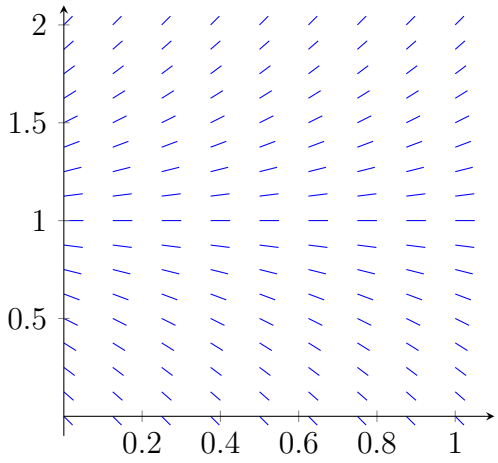
$$\frac{dy}{dx} = ____.$$

Fill in the blanks: For the equation $\frac{dy}{dt} = g(t, y)$, the equilibria are the ____-values where _____ (or $g(t, y) = 0$). On the slopefield, they will correspond to a y value at which we see only _____ ticks.

Definition An equilibrium is called *stable* if a small change in the initial condition gives a solution which tends back to the equilibrium at $t \to \infty$. An equilibrium is *unstable* if a small change in the initial condition gives a solution which veers away from the equilibrium as $t \to \infty$.

1. Consider the differential equation we saw last time, $\frac{dy}{dx} = 2(y-1)$. Here is its slopefield again:

□Regrade

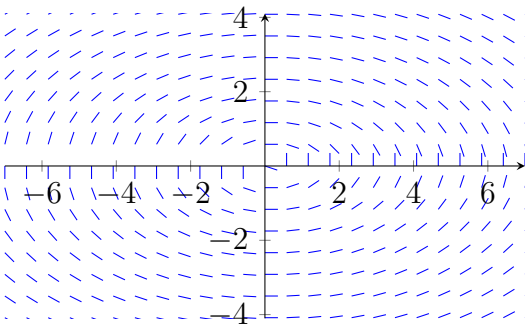


(a) What is its equilibrium? Draw the equilibrium solution on the slopefield.

(b) Draw solutions on the slopefield. Is the equilibrium stable or unstable?

2. Consider the equation $\frac{dy}{dt} = -\frac{t}{3y}$. Here is its slopefield again:

□Regrade

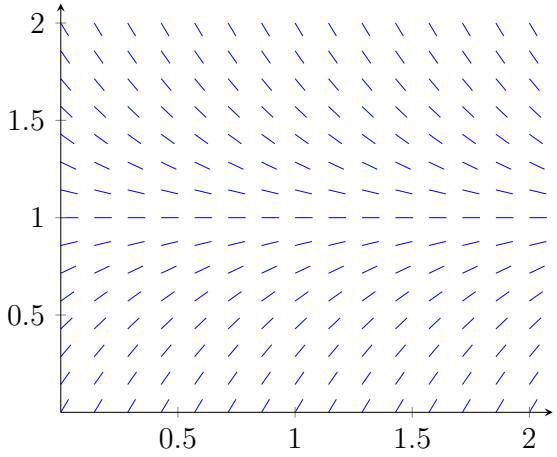


(a) Looking at the slopefield, why does it appear this equation has no equilibria?

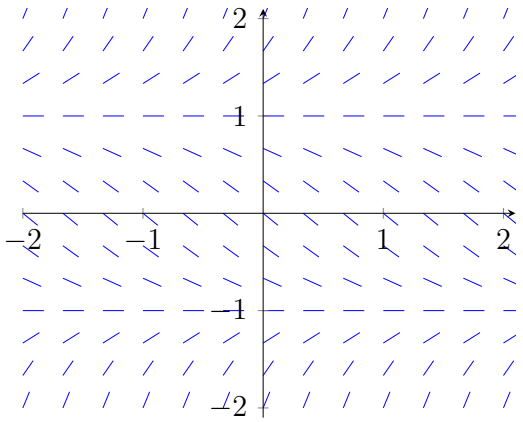
(b) Use the equation to prove this.

3. For each of the differential equations below, find their equilibrium solutions, then use □Regrade their slopefields to determine if each equilibrium is stable or unstable.

(a) $\frac{dy}{dt} = -2(y-1)$. (Compare this to Question 1 above!)



(b) $\frac{dy}{dt} = (y-1)(y+1)$.



(c) $\frac{dy}{dt} = ty(t-1)(y-1)(y+1)$.

